Fiscal Year:	FY 2005	**Task Last Updated:**	FY 10/21/2005
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Project Title:	Circadian Entrainment, Sleep-Wake Regulation and Neurobehavioral Performance During Extended Duration Space Flight		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Human Performance Factors, Sleep, and Chronobiology Team		
Joint Agency Name:		**TechPort:**	No
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Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02115-5804	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
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No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
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COI Name (Institution):			
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Task Description:	NASA's planned exploration class mission to Mars will require crewmembers to adapt to a 24.65-h solar day throughout their >540 day stay on Mars. The ability of astronauts to sustain a high level of performance during their mission will be critically dependent upon adaptation of their circadian pacemaker to the 24.65-h day. Our data suggest that most astronauts would exhibit circadian misalignment if the space flight lighting conditions of <25 lux on the windowless middeck of the space shuttle were present during their stay on Mars. Circadian misalignment would result in sleep impairment, endocrine disturbances, and impaired neurobehavioral function. Preliminary data from our laboratory reveal that intermittent bright light exposure is effective in maintaining entrainment of the circadian pacemaker to longer-than-24 h days. They also suggest that lighting levels of ~25 lux or ~100 lux are insufficient to maintain a normal phase angle of entrainment. Other preliminary results demonstrate that blue light (460 nm) is more efficient than white light in resetting the human circadian pacemaker. Given that bright light pulses might not of practical use on Mars—due to time and energy constraints—these results demonstrate the need to develop practical and cost-effective countermeasures to adapt the human circadian pacemaker of all crewmembers to the 24.65-h day. We propose to test a countermeasure of "blue-enriched" light to entrain the human circadian pacemaker to a 24.65-h day. Specifically, we propose to test the following hypotheses: i) that synchronization of the human circadian pacemaker to a 24.65-h day initiated at an adverse phase will not be appropriate in the presence of ~100 lux of white light; ii) that inappropriate circadian synchronization will result in the secretion of the sleep-promoting hormone melatonin during waketime, abnormal somatotropic and corticotropic activity, disturbed sleep, and impaired performance and daytime alertness; and iii) that exposure to "blue enriched" light during the daytime will establish a normal entrained circadian phase in subjects scheduled to a 24.65-h day. We have proposed a randomized 71-day clinical trial to test these three hypotheses. The results of the proposed studies will answer fundamental questions on the mechanisms underlying circadian entrainment in humans and could have a profound effect on the health, productivity and safety of astronauts during an exploration class mission to Mars.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The purpose of the proposed studies is to test three specific hypotheses aimed at evaluating the effectiveness of blue-enriched light exposure over white light as a countermeasure for circadian misalignment during exploration class mission to mars. Results from previous studies in our lab indicate exposure to blue-enriched light will be a more effective means of entraining the human circadian pacemaker compared to exposure to white light of similar photon density, or even to light of greater photon density but of inappropriate spectral content (long wavelengths, as on Mars). The funded research project has important implications for earth-based applications, particularly in the development of treatment for circadian rhythm disorders. Since the finding that appropriately scheduled light/dark exposure is a powerful means of resetting the human circadian pacemaker, bright light has been used in various clinical settings to induce physiologic adaptation in individuals suffering from circadian rhythm disorders (e.g. night shift workers, people with Advanced and Delayed Sleep Phase Syndromes). Misalignment of circadian phase and work-sleep schedules can overpower the night-shift worker's ability to remain awake and attentive while at work leading to impaired job performance and higher rates of accidents and injuries. We have found that appropriately timed exposure to bright light can produce rapid physiologic adaptation of the circadian pacemaker to a single week of night work and facilitate rapid entrainment to a rotating work schedule, as well as enhance the alertness of night workers during their work shifts. Advanced and Delayed Sleep Phase Syndromes (ASPS and DSPS, respectively) are characterized by a marked difficulty in maintaining appropriate timing of sleep during the desired hours, and there is evidence suggesting that circadian misalignment may underlie the pathophysiology of this condition. We and others have reported data from clinical studies which suggest that evening exposure to bright light or early morning exposure to bright light are successful in the treatment of Advanced Sleep Phase Syndrome or Delayed Sleep Phase Syndrome, respectively. Since the current study is designed to evaluate the effectiveness of blue-enriched light exposure over white light as a countermeasure for circadian misalignment during exploration class mission to mars, it has implications for the development of effective and energy-efficient methods for treatment of circadian rhythm disorders. Optimizing the wavelength of light holds the potential for producing shorter, more efficient light treatment regimens. Shorter treatment regimens would not only increase compliance in clinical populations, but would make light treatment more practical in industrial/work settings.
Task Progress:	We have completed two subjects in the proposed 71-Day protocol, one over the last funded year and one this year. A third subject is being studied currently. Two more subjects are scheduled to start the study in June-july. Data collected in the two studies include: Core body temperature, blood samples (melatonin, cortisol, IGF-1), Sleep and waking EEG recordings, Subjective sleep quality, Actigraphy, Light intensity, neurobehavioral performance and mood. The successful collection of these data will allow us to test hypotheses 1, 2, and 3 of the project. Analyses of the collected data are currently in progress.
Bibliography Type:	Description: (Last Updated: 11/17/2025)
Articles in Peer-reviewed Journals	Hull JT, Wright KP Jr, Czeisler CA. "The influence of subjective alertness and motivation on human performance independent of circadian and homeostatic regulation." J Biol Rhythms. 2003 Aug;18(4):329-38. PMID: 12932085 , Jan-2004
Articles in Peer-reviewed Journals	Wright Jr, K. P., Gronfier, C, Duffy, J. F., Czeisler, C. A. "Intrinsic circadian period and strength of the circadian synchronizer determines the phase relationship between melatonin onset, habitual sleep time and light-dark cycles in humans." N/A , Jan-2004
Articles in Peer-reviewed Journals	Wright KP Jr, Gronfier C, Duffy JF, Czeisler CA. "Intrinsic period and light intensity determine the phase relationship between melatonin and sleep in humans." J Biol Rhythms. 2005 Apr;20(2):168-77. PMID: 15834113 , Apr-2005
Presentation	Gronfier, C "Entrainment of the human circadian timing system to a longer-than-24-h sleep-wake cycle" N/A Jun-2004
Presentation	Gronfier, C "Entrainment of the human circadian timing system to a longer-than-24-h sleep-wake cycle." N/A May-2004
Presentation	Gronfier, C "Use of intermittent bright light for entrainment of the human circadian pacemaker." N/A Jul-2004

Presentation	Gronfier, C. "Circadian entrainment, sleep-wake regulation and neurobehavioral performance during space flight" N/A Jan-2004
Presentation	Wright, K. P. Jr. "Control of Waking Functions by Homeostatic and Circadian Processes." N/A Jun-2004
Presentation	Wright, K. P. Jr. "Entrainment of the Human Biological Clock" N/A Feb-2004
Presentation	Wright, K. P. Jr. "Photic entrainment of the human circadian timing system to the day length of Mars" N/A Jul-2004
Presentation	Wright, K. P. Jr. "Principles of Circadian Entrainment in Humans" N/A May-2004